```c
#include <stdio.h>
int main(){
  int a[3][3]={{3,6,7,},{4,6,9},{2,5,8}};
  int b[3][3]={{2,5,-9},{5,4,6},{8,6,7}};
  int c[3][3],i,j;
  for(i=0;i<3;i++)
  {
    for(j=0;j<3;j++)
    {
      c[i][j]=a[i][j]+b[i][j];
      printf("%d ",c[i][j]);
    }
    printf("\n");
  }
    return 0;
}
```

```c
#include <stdio.h>
int main(){
  int a[100],n,i,ele;
  scanf("%d",&n);
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  scanf("%d",&ele);
  for(i=n-1;i>=0 && a[i]>ele;i--)
    a[i+1]=a[i];
  a[i+1]=ele;
  n++;
  for(i=0;i<n;i++)
    printf("%d ",a[i]);
  return 0;
}
```

```c
#include <stdio.h>
int main(){
  int a[100],n,i,j,key;
  scanf("%d",&n);
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  scanf("%d",&key);
  for(i=0;i<n;i++)
    if(a[i]==key){
      for(j=i;j<n-1;j++)
        a[j]=a[j+1];
      n--;
      i--;

    }
       for(i=0;i<n;i++)
       printf("%d ",a[i]);
       return 0;
}
```

```c
#include <stdio.h>
int main(){
  int a[100],n,i,flag=1;
  scanf("%d",&n);
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  for(i=0;i<n/2;i++){
    if(a[i]!=a[n-1-i]){
      flag=0;
    break;
    }
  }
  if(flag){
    printf("Palindrome");
  }
    else
        printf("Not palindrome");
    return 0;
  }
```

```c
#include <stdio.h>
int main(){
  int a[100],b[100],n1,n2,i,j;
  scanf("%d",&n1);
  for(i=0;i<n1;i++)
    scanf("%d",&a[i]);
  scanf("%d",&n2);
  for(i=0;i<n2;i++)
    scanf("%d",&b[i]);
  for(i=0;i<n1;i++)
    for(j=0;j<n2;j++)
      if(a[i]==b[j])
        printf("%d",a[i]);
  return 0;
}
```

```c
#include <stdio.h>
int main()
{
  int a[3][3]={{3,6,7},{4,6,9},{2,5,8}};
  int i,sum=0;
  for(i=0;i<3;i++)
    sum+=a[i][i];
  printf("sum of diagonal =%d",sum);
  return 0;
}
```

```c
#include <stdio.h>
int main(){
  int a[3][3],b[3][3],c[3][3];
  int i,j,k;
  for(i=0;i<3;i++)
    for(j=0;j<3;j++)
      scanf("%d",&a[i][j]);
  for(i=0;i<3;i++)
    for(j=0;j<3;j++)
      scanf("%d",&b[i][j]);
  for(i=0;i<3;i++){
    for(j=0;j<3;j++){
      c[i][j]=0;
  for(k=0;k<3;k++)
    c[i][j]+=a[i][k]*b[j][k];
    }
  }
  for(i=0;i<3;i++){
    for(j=0;j<3;j++)
      printf("%d ",c[i][j]);
  printf("\n");
```

```c
#include <stdio.h>

int main() {
    int r, c, i, j;
    int matrix[10][10], transpose[10][10];

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("\nEnter matrix elements:\n");
    for (i = 0; i < r; ++i) {
        for (j = 0; j < c; ++j) {
            printf("Enter element a%d%d: ", i + 1, j + 1);
            scanf("%d", &matrix[i][j]);
        }
    }


    for (i = 0; i < r; ++i) {
        for (j = 0; j < c; ++j) {
            transpose[j][i] = matrix[i][j];
```

```c
#include <stdio.h>

int main() {
    int rows, cols, i, j;
    int matrix[100][100];
    int sum = 0;

    printf("Enter the number of rows (max 100): ");
    if (scanf("%d", &rows) != 1) return 1;

    printf("Enter the number of columns: ");
    if (scanf("%d", &cols) != 1) return 1;

    printf("Enter the elements of the matrix:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    printf("The matrix is:\n");
```

```c
#include <stdio.h>

int main() {
    int n, i, j;
    int matrix[10][10];
    int product = 1;

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    for (i = 0; i < n; i++) {
        product *= matrix[i][i];
    }

```

```c
#include <stdio.h>

int main() {
    int n, i, j, sum = 0;
    int matrix[10][10];

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    for (i = 0; i < n; i++) {
        for (j = i; j < n; j++) {
            sum += matrix[i][j];
        }
    }
```

```c
#include <stdio.h>

int main() {
    int n, i, j, sum = 0;
    int matrix[10][10];

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    for (i = 0; i < n; i++) {
        for (j = 0; j <= i; j++) {
            sum += matrix[i][j];
        }
    }
```

```c
#include <stdio.h>

int main() {
    int n, i, j, sum = 0;
    int matrix[10][10];

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    // Sum of left diagonal elements
    for (i = 0; i < n; i++) {
        sum += matrix[i][i];
    }

```

```c
#include <stdio.h>

int main() {
    int n, i, j, sum = 0;
    int matrix[10][10];

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    for (i = 0; i < n; i++) {
        sum += matrix[i][n - 1 - i];
    }

    printf("Sum of right diagonal elements = %d\n", sum);
```

```c
#include <stdio.h>

int main() {
    int n, i, j, symmetric = 1;
    int matrix[10][10];

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            if (matrix[i][j] != matrix[j][i]) {
                symmetric = 0;
                break;
```

```c
#include <stdio.h>

int main() {
    int n, i, j, diagonal = 1;
    int matrix[10][10];

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            if (i != j && matrix[i][j] != 0) {
                diagonal = 0;
                break;
```

```c
#include <stdio.h>

int main() {
    int n, i, j, upper = 1;
    int matrix[10][10];

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    for (i = 1; i < n; i++) {
        for (j = 0; j < i; j++) {
            if (matrix[i][j] != 0) {
                upper = 0;
                break;
```

```c
#include <stdio.h>

int main() {
    int n, i, j, lower = 1;
    int matrix[10][10];

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (matrix[i][j] != 0) {
                lower = 0;
                break;
```

```c
#include <stdio.h>

int main() {
    int r, c, i, j, equal = 1;
    int a[10][10], b[10][10];

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter first matrix:\n");
    for (i = 0; i < r; i++) {
        for (j = 0; j < c; j++) {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter second matrix:\n");
    for (i = 0; i < r; i++) {
        for (j = 0; j < c; j++) {
            scanf("%d", &b[i][j]);
        }
```

```c
#include <stdio.h>

int main() {
    int r, c, i, j;
    int a[10][10], b[10][10], result[10][10];

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter first matrix:\n");
    for (i = 0; i < r; i++) {
        for (j = 0; j < c; j++) {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter second matrix:\n");
    for (i = 0; i < r; i++) {
        for (j = 0; j < c; j++) {
            scanf("%d", &b[i][j]);
        }
```

```c
#include <stdio.h>

int main() {
    int r, c, scalar, i, j;
    int matrix[10][10];

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter matrix elements:\n");
    for (i = 0; i < r; i++) {
        for (j = 0; j < c; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    printf("Enter scalar: ");
    scanf("%d", &scalar);

    for (i = 0; i < r; i++) {
        for (j = 0; j < c; j++) {
```

```c
#include <stdio.h>

int main() {
    int a[2][2], determinant;

    printf("Enter 2x2 matrix elements:\n");

    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            scanf("%d", &a[i][j]);
        }
    }

    determinant = (a[0][0] * a[1][1]) - (a[0][1] * a[1][0]);

    printf("Determinant = %d\n", determinant);

    return 0;
}
```

```c
#include <stdio.h>

int main() {
    float a, b, c, d;
    float det;

    printf("Enter elements of 2x2 matrix:\n");
    scanf("%f %f %f %f", &a, &b, &c, &d);

    det = a * d - b * c;

    if (det == 0) {
        printf("Inverse does not exist.\n");
    } else {
        printf("Inverse of the matrix:\n");
        printf("%.2f %.2f\n", d / det, -b / det);
        printf("%.2f %.2f\n", -c / det, a / det);
    }

    return 0;
}
```

```c
#include <stdio.h>

int main() {
    int a[10][10], m, n, i, j, k, rank = 0;
    float ratio;

    printf("Enter rows and columns: ");
    scanf("%d %d", &m, &n);

    printf("Enter matrix elements:\n");
    for (i = 0; i < m; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &a[i][j]);

    for (j = 0; j < n && rank < m; j++) {
        for (i = rank; i < m; i++) {
            if (a[i][j] != 0) {
                for (k = 0; k < n; k++) {
                    int temp = a[rank][k];
                    a[rank][k] = a[i][k];
                    a[i][k] = temp;
```

```c
#include <stdio.h>

int main() {
    int a[10][10], r, c, i, j, k;
    float ratio;

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter matrix elements:\n");
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &a[i][j]);

    for (i = 0; i < r; i++) {

        if (a[i][i] == 0) {
            for (k = i + 1; k < r; k++) {
                if (a[k][i] != 0) {
                    for (j = 0; j < c; j++) {
                        int temp = a[i][j];
```